Mutually Exclusive Events and Either/Or

Name _____ Date _____ No calculator Version 1

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KEY WORDS event *(takahanga)* · mutually exclusive · either/or · outcome *(putanga)*

A Can both happen? ●○○

Remember: Two events are mutually exclusive if they cannot both happen at the same time. Rolling a 2 and rolling a 5 on one dice are mutually exclusive.

1. An ordinary dice is rolled once. Are the two events mutually exclusive?

a) 2 and 5

b) Even number and 3

c) Even number and prime number

d) Odd number and even number

e) Number greater than 4 and multiple of 3

f) Number less than 3 and number greater than 4

g) Square number and odd number

h) Multiple of 3 and 4

i) Prime number and 1

j) Number greater than 2 and number less than 5

k) Multiple of 4 and odd number

l) Multiple of 3 and even number

B Adding probabilities ●●○

Remember: If two events are mutually exclusive, add their probabilities to find the probability of one or the other.

2. An ordinary dice is rolled. Find the probability of getting:

a) a 2 or a 5

b) an even number or a 3

c) a number less than 3 or a number greater than 4

d) a multiple of 3 or a 4

e) a prime number or a 1

f) a multiple of 4 or an odd number

3. Cards numbered 1 to 10 are shuffled and one card is picked. Find the probability of getting:

a) a number less than 3 or a number greater than 8

b) a multiple of 5 or a 7

c) a multiple of 4 or a multiple of 3

d) a square number or a 10

e) a prime number or a multiple of 4

f) a multiple of 5 or a multiple of 3

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C When events overlap ●●●

⚠ **Watch out:** If an outcome fits both events, adding counts it twice. Take away the probability of both happening.

4. A spinner has 8 equal sections numbered 1 to 8. Find the probability that it lands on:

a) an even number or a number greater than 5

b) a prime number or an odd number

c) a multiple of 4 or an even number

d) a square number or a number less than 3

5. An ordinary dice is rolled. Find the probability of getting:

a) an even number or a prime number

b) a number greater than 3 or an even number

c) an odd number or a multiple of 3

d) a square number or a number less than 3

6. Cards numbered 1 to 12 are shuffled and one card is picked. Find the probability of getting:

a) an even number or a multiple of 3

b) a prime number or an odd number

c) a multiple of 4 or an even number

d) a number greater than 8 or a multiple of 5

D Problems ●●●

7. In a class of 30 students, 12 play football and 10 play netball. 4 of these students play both. One student is chosen at random. Find the probability that the student plays football or netball.

8. A bag holds 5 red, 3 blue and 4 green counters. One counter is picked. Find the probability that it is red or green.

9. The probability that a bus is early is 0.15, and the probability that it is on time is 0.6. Find the probability that it is early or on time.

★ Extension

10. Cards numbered 1 to 20 are shuffled and one is picked. Find the probability of getting an even number or a multiple of 5. First list the numbers that are both.

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Mutually Exclusive Events and Either/Or

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS event *(takahanga)* · mutually exclusive · either/or · outcome *(putanga)*

A Can both happen? ●○○

Remember: Two events are mutually exclusive if they cannot both happen at the same time. Rolling a 2 and rolling a 5 on one dice are mutually exclusive.

1. An ordinary dice is rolled once. Are the two events mutually exclusive?

a) 1 and 6

b) Odd number and 4

c) Odd number and prime number

d) Even number and odd number

e) Number less than 3 and even number

f) Number less than 2 and number greater than 5

g) Square number and even number

h) Multiple of 3 and 5

i) Prime number and 4

j) Number greater than 3 and number less than 6

k) Multiple of 4 and prime number

l) Multiple of 3 and odd number

B Adding probabilities ●●○

Remember: If two events are mutually exclusive, add their probabilities to find the probability of one or the other.

2. An ordinary dice is rolled. Find the probability of getting:

a) a 1 or a 6

b) an odd number or a 4

c) a number less than 2 or a number greater than 4

d) a multiple of 3 or a 5

e) a prime number or a 4

f) a multiple of 4 or a prime number

3. Cards numbered 1 to 10 are shuffled and one card is picked. Find the probability of getting:

a) a number less than 4 or a number greater than 9

b) a multiple of 5 or a 3

c) a multiple of 4 or a multiple of 3

d) a square number or a 8

e) a prime number or a multiple of 4

f) a multiple of 5 or a multiple of 3

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C When events overlap ●●●

> ⚠ **Watch out:** If an outcome fits both events, adding counts it twice. Take away the probability of both happening.

4. A spinner has 8 equal sections numbered 1 to 8. Find the probability that it lands on:

a) an odd number or a number greater than 4

b) a prime number or an even number

c) a multiple of 3 or an odd number

d) a square number or a number less than 5

5. An ordinary dice is rolled. Find the probability of getting:

a) an odd number or a prime number

b) a number greater than 2 or an even number

c) an even number or a multiple of 3

d) a square number or a number less than 4

6. Cards numbered 1 to 12 are shuffled and one card is picked. Find the probability of getting:

a) an even number or a multiple of 4

b) a prime number or a number less than 6

c) a multiple of 3 or an odd number

d) a number greater than 7 or a multiple of 4

D Problems ●●●

7. In a class of 40 students, 18 play football and 15 play netball. 5 of these students play both. One student is chosen at random. Find the probability that the student plays football or netball.

8. A bag holds 6 red, 5 blue and 4 green counters. One counter is picked. Find the probability that it is red or green.

9. The probability that a bus is early is 0.15, and the probability that it is on time is 0.6. Find the probability that it is early or on time.

★ Extension

10. Cards numbered 1 to 20 are shuffled and one is picked. Find the probability of getting a multiple of 3 or a multiple of 4. First list the numbers that are both.

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